

Lab Task 04

Use-case Model and Project Scheduling

1. Use-case diagram for our project

- List the actors, Use cases and relationships.

Primary Actors

1. User

Core Use Cases

1. Daily mood check in and showing them in the calendar.
2. Writing text journals and storing them in the database.
3. Chatting with an AI chatbot to provoke mindful conversations.
4. Enter contact information.
5. Import/Export User Data.

Key Relationships

1. User — communicates → Onboarding
2. Contact Information — <<extends>> → Onboarding
3. User — communicates → Mood Check-in
4. Mood Check-in — <<includes>> → Calendar
5. User — communicates → Text Journal
6. Text Journal — <<includes>> → Calendar
7. User — communicates → Chat Interface
8. User — communicates → Import/Export User Data

- Draw the use case diagram using tool: **draw.io**.

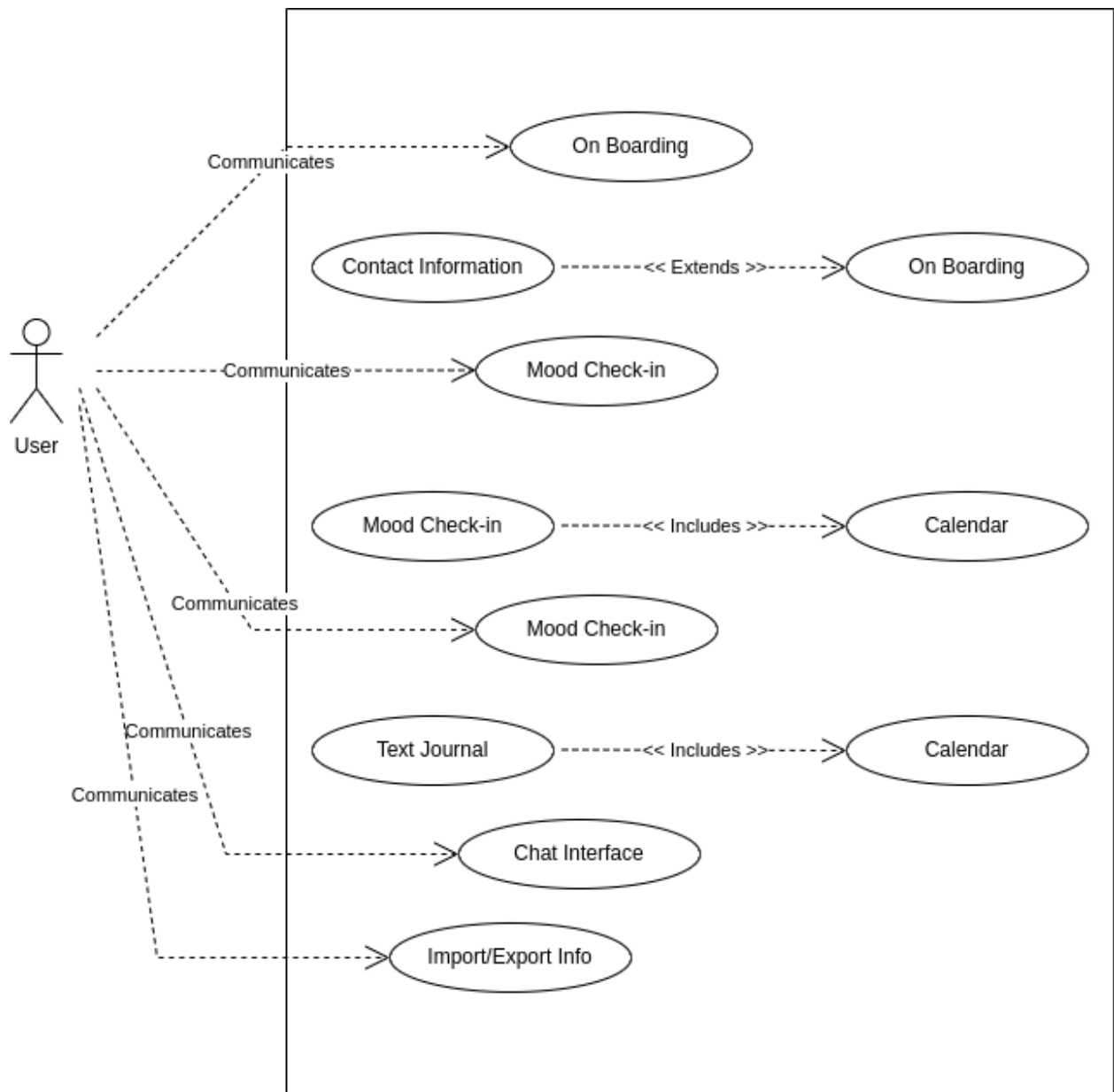


Fig. 1: Use case diagram of our proposed project.

- For each use case also describe the steps needed to perform before the use case.

1. Onboarding

- *Basic*: Onboarding → Mood Check-in
- *Alternate*: Onboarding → Contact Information → Mood Check-in

2. Mood Check-in

- *Basic*: Homescreen → Mood Check-in → Calendar → Add Day Entry → Review previous entries
- *Alternate*: Onboarding → Mood Check-in → Calendar → Add Day Entry

3. Text Journal

- *Basic*: Homescreen → Text Journal
- *Alternate*: Homescreen → Text Journal → Edit Journal Card

4. Chat Interface

- *Basic*: Homescreen → Chat Interface

5. Import/Export Information

- *Basic*: Homescreen → Settings → Import/Export Information

2. Describe the design methodology you are using for your project (SDLC, Agile, or OOSAD). Why is the selected methodology best suited for your project?

We are using the Agile methodology for our project among SDLC, Agile or OOSAD. We think the selected methodology (Agile) is best suited for our project because Agile is a flexible framework, allowing rapid changes and providing incremental updates. We can add new features while we solidify the core features and postpone the high risk features until they are ready to be implemented.

The phases of Agile:

- Exploration
- Planning
- Iterations of the first release
- Productionizing
- Maintenance

Since Agile is a user-centric development framework, we can divide the work in short-cycles, conduct daily meetings, test software and collaborate as a team.

3. Schedule your project using Gantt Chart.

- To create a Gantt chart, start by breaking down your project into phases, then into activities, and finally into detailed tasks. Assign a duration to each task and identify its predecessor: the task that must be completed before it begins.

Organize this information into a table with columns for activity, predecessor, and duration. Using this structured data, draw the Gantt chart to visualize the project timeline effectively.

A	Conduct interviews
B	Administer questionnaires
C	Read company reports
D	Analyze data flows
E	Introduce prototype
F	Observe reactions
G	Perform cost-benefit
H	Prepare proposal
I	Present proposal

Phases	Detailed Activity	Predecessor	Duration (Weeks)
A	1. Conduct interviews. 2. Identify requirements. 3. Identify weaknesses of existing solutions.	-	1
B	1. Prepare relevant questions. 2. Invite stakeholders for an interview. 3. Perform and analyze interviews.	A	1
C	1. Read company reports. 2. Identify and allocate available human resources.	B	1
D	1. Draw Context-Level DFD and E-R diagrams. 2. Draw Use case diagram.	C	1
E	1. Create a working prototype with only the essential features. 2. Build and release the prototype for public Beta use.	D	2

F	1. Observe user feedback and overall perception of the software. 2. Perform A/B when enough data is available during its early adoption.	E	1
G	1. Reevaluate the cost of building the working prototype and how the costs would increase over time. 2. Allocated required budget.	F	1
H	1. Prepare a proposal highlighting the user feedback and reactions during the prototype build.	G	2
I	1. Perform a presentation in a tech convention, proposing the software. 2. Raise fund (if necessary) to further facilitate the development of the proposed software.	H	1

- Draw the Gantt Chart.

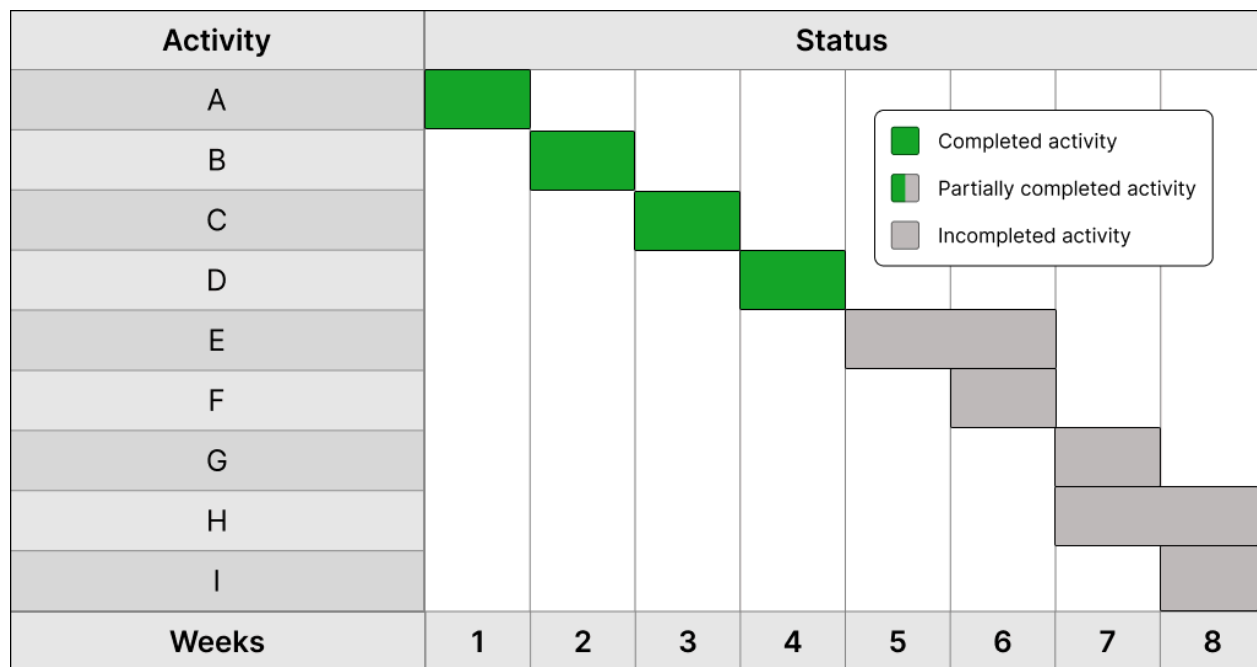


Fig: 2: Gantt chart of our proposed project.